



How and when does boundary spanning behavior influence employees' creative performance? The roles of meaning of work and team Chaxu climate

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Abstract

With the increasing uncertainty of the environment and the complexity of team tasks, it's now common practice to encourage employees to implement boundary spanning behaviors. In spite of the fact that a few studies have explored the relationship between boundary spanning behavior and employee creative performance, little is known about the mechanism behind this association. Drawing on the conservation of resource theory (COR), this paper examines the mediating role of meaning of work and the moderating role of team Chaxu climate to reveal the internal process and boundary condition regarding the effect of boundary spanning behavior on employee creative performance. A sample of 211 supervisor-subordinate dyads from six companies located in China is collected in this study, and a hierarchical linear model is used to test the hypotheses. The results show that boundary spanning behavior has a positive effect on employee creative performance, meaning of work mediates the link between them, and team Chaxu climate negatively moderates the relationship between boundary spanning behavior and meaning of work. This research provides a new insight into the linkage between boundary spanning behavior and employee creative performance, as well as further answers the question “how and when boundary spanning behavior influences employee creative performance”.

Keywords Boundary spanning behavior · Creative performance · Meaning of work · Team Chaxu climate

Introduction

With the increasing uncertainty of the environment and the complexity of team tasks, many leaders encourage their members to cross team boundaries and establish connections with external parties through boundary spanning behaviors to achieve team goals (Liu et al., 2018). Boundary Spanning Behavior (BSB) refers to the behavior aimed at establishing relationships and interactions with external actors to achieve the overall team goals (Marrone et al., 2007). Studies have shown that employees' BSB can help to

improve team performance (Marrone et al., 2007) and organizational innovation (Hargadon, 1998). As research continues to advance, scholars have also begun to focus on how BSB affects people individually—particularly in the area of individual innovation. For example, based on knowledge integration theory and status characteristics theory, existing studies proposed that employees' BSB can improve their creative performance through integrating internal and external knowledge and obtaining informal status (Zhang & Li, 2020a; Zhang & Li, 2023b). However, research on the mechanism of the relationship between BSB and employee creative performance needs to be further explored (Zhang & Li, 2020a). This is because both knowledge and informal status are the resources acquired by individuals in the process of establishing links with external parties. Individuals can also acquire other heterogeneous resources in this process and need to invest certain resources to achieve creative outcomes (Liu et al., 2018; Marrone, 2010). That is, BSB affects employees' creative performance through the acquisition, investment, and transformation of various resources.

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However, previous research focused on specific resources such as knowledge and informal status, which does not fully reveal the above processes. This makes it difficult to provide a strong response to the question of how and when the resources that employees receive through BSB are transformed into creative performance within the individual. The COR theory provides a systematic framework for interpreting the dynamic change of individual resources (Liao et al., 2022). Therefore, it is necessary to examine the mechanism of BSB on creative performance based on COR theory.

People can acquire a variety of resources with the aid of BSB, such as knowledge, informal status, and social relationships (Liu et al., 2018; Zhang & Li, 2020a; Zhang & Li, 2023b). So how do these resources translate into employees' creative performance? According to COR theory, people must invest resources to gain more resources. Meanwhile, those who have an abundance of resources have greater opportunities and capacities to get more resources, leading to a spiral gain in resources (Hobfoll et al., 2018). Therefore, if the boundary spanners want to achieve high creative performance, they must invest resources in creative activities. However, creative activities require significant resources and entail high risks, people must be sufficiently intrinsically motivated to engage in them (Wang & Kou, 2023). The boundary spanners are capable of assisting their team in achieving goals. This increases their sense of the worth and significance of their work, and then enhances their meaning of work (Lysova et al., 2019; Rosso et al., 2010). As a major source of intrinsic motivation for individuals, meaning of work helps to promote the boundary spanners to invest their resources into creative activities to achieve the spiral growth of resources and in turn improve their creative performance (Allan et al., 2019; Cohen-Meitar et al., 2009). Thus, this research introduces meaning of work as a mediating variable to reveal the trans-formative process of the resources that individuals gain from BSB.

In addition, the resource caravan passageways principle states that individuals' creation and sustenance of the resources are influenced by the environment. Organizations and the broader culture play a major role in this process

(Hobfoll et al., 2018). Considering the widespread existence of phenomena such as “quanzi” and circles in organizations that have a profound impact on individual behavior and the allocation of organizational resources under the cultural characteristics of China (Liu et al., 2009), we intend to investigate the influence of team Chaxu climate on the relationship between the BSB and meaning of work. Team Chaxu climate as an essential phenomenon of interpersonal relationships within organizations based on traditional Chinese culture and social psychology, reflects the distribution characteristics of team resources and power (Chen & Dian, 2018). When the Chaxu climate of a team is stronger, team members are more likely to perceive the team's injustice for the distribution of resources and get much passive emotion (Chen & Dian, 2018), which is not conducive to the formation of their psychological resource (i.e., meaning of work). Thus, we argue that team Chaxu climate may negatively affect the relationship between BSB and meaning of work.

In summary, based on COR theory, this study introduces meaning of work as a mediating variable and team Chaxu climate as a moderating variable to further reveal the internal process and boundary conditions regarding the effect of BSB on employees' creative outcomes. It not only enriches the research on the relationship mechanism between BSB and employee creative performance from a new perspective, but also provides guidance and reference for organizational management practice. The theoretical model is displayed in Fig. 1.

Literature review and hypotheses

Conservation of resource theory (COR)

Conservation of resource theory states that individuals strive to obtain, retain, foster, and protect those things they consider valuable (Hobfoll, 1989). It includes multiple principles and corollaries, such as resource investment principle, resource caravan passageways principle, and resource gain/loss spirals. Resource investment principle is that

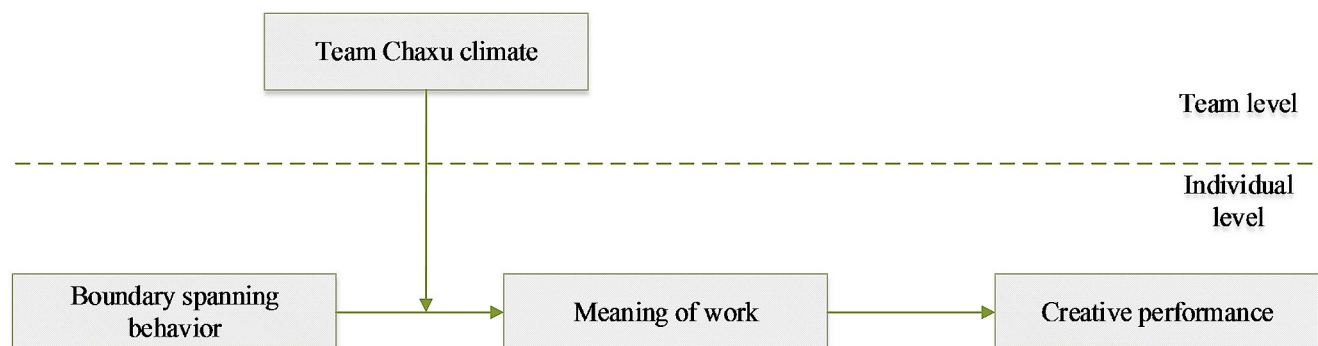


Fig. 1 Theoretical Model

people must invest resources in order to prevent resource loss and gain resources (Hobfoll, 1989). Resource caravan passageways principle means that resources exist in ecological conditions that either foster or limit resource creation and sustenance (Hobfoll, 2011). These views provide a systematic framework for interpreting the dynamic change of individual resources (Liao et al., 2022).

BSB and employee creative performance

Employee creative performance refers to the novel or original products, ideas and procedures at the individual level that benefit organizations (Oldham & Cummings, 1996). Previous research has attempted to identify the individual or contextual factors affecting employee creative performance (Shalley et al., 2004; Wadei et al., 2023; Abukhait et al., 2023). For example, related literature has shown that employees' feedback-seeking behaviors or team external learning behaviors can facilitate individuals' creative performance (Liu et al., 2013). BSB is different from the above constructs, but it has some similar characteristics to them. For example, BSB also includes information searching and sharing, seeking feedback from top managers and learning from external interactions (Liu et al., 2018; Marrone et al., 2007). We argue that there may be a positive relationship between BSB and creative performance.

First, COR theory points out that individuals will invest in getting more valuable resources to enrich their resource pools when they have enough resources (Hobfoll, 1989). Boundary spanners can get various resources from internal or external stakeholders through BSB (Marrone, 2010). Therefore, they are more likely to use these resources to carry out creative activities to achieve higher creative performance. This helps to resist the loss of resources and meets the expectations of team leaders for boundary spanners (Zhang & Li, 2020a).

Second, boundary spanners can establish good relationships with outsiders when engaging in BSB, such as customers, executives and professionals. This can enable them to occupy the "structural holes" (Marrone, 2010). The actors in these key positions can acquire knowledge, information, and other new resources, which can help them generate creative ideas and obtain more creative performance (Zhang & Li, 2020a). Moreover, employees in the center of the network have the advantage of exchanging heterogeneous resources (e.g., knowledge or information) with their external stakeholders and establishing high-quality team-member exchange (TMX) relationships with the other members of their teams (Marrone, 2010). Previous research has shown that heterogeneous knowledge and TMX are positively related to employee creative performance (Liao et al., 2010).

Finally, the implementation of creative ideas has certain risks and challenges. Individuals will face the threat of resource loss in this process (Yuan & Woodman, 2010). According to COR theory, those with greater resources are not only less vulnerable to resource loss but also more capable of resource gain (Hobfoll et al., 2018). This means that boundary spanners who acquire more resources can well cope with the uncertainties in creative activities and promote the effective implementation of creative ideas, ultimately leading to high creative performance. In summary, BSB can not only enhance employees' creative willingness but also promote the generation and implementation of their creative ideas. Therefore, we propose:

H1 BSB is positively related to employee creative performance.

The mediating role of meaning of work

Meaning of work refers to individuals' understanding of what they do at work and the significance of what they do (Wrzesniewski et al., 2003). It comes from many aspects such as individual characteristics, leadership, and work context (Bailey et al., 2019; Rosso et al., 2010), and helps individuals produce positive work attitudes and behaviors (Song et al., 2018). According to COR theory, people must invest resources to gain resources. Meanwhile, individuals need to have sufficient intrinsic motivation to invest in creative activities that are characterized by high risk and high investment (Hobfoll et al., 2018; Wang & Kou, 2023). As a major source of intrinsic motivation, meaning of work helps to promote the boundary spanners to invest their resources into creative activities to achieve the spiral growth of resources and in turn improve their creative performance (Duan et al., 2023). Therefore, this research introduces meaning of work as a mediating variable.

Boundary spanning behavior may impact employees' meaning of work positively. First, research has shown that individuals will experience meaning of work when their work roles are purposeful and worthwhile (Clausen & Borg, 2011). Boundary spanners take more responsibility for establishing contact with outsiders and acquiring resources to achieve team goals. Thus, they will perceive more value and importance and then experience more meaning of work (Marrone et al., 2007). Second, COR theory states that condition resources can create conditions for individuals to acquire other key resources (Hobfoll, 1989). BSB is helpful in acquiring condition resources for individuals, such as high-quality workplace relationships, informal leadership and influence (Marrone, 2010; Liu et al., 2018; Marrone et al., 2007), which may provide support for the formation

of meaning of work (Clausen & Borg, 2011; Bailey et al., 2019). Third, according to COR theory (Hobfoll, 1989), individuals invest to gain new resources and protect against resource loss. As a kind of resource investing behavior, BSB provides a good resource base for employees to perceive the meaning of work (Song et al., 2021). Accordingly, we speculate that BSB may help individuals to get meaning of work.

Furthermore, we argue that meaning of work is positively related to employee creative performance. On the one hand, individuals who experience more meaning of work have abundant psychological resources (Clausen & Borg, 2011; Sun et al., 2019). According to COR theory, those with greater resources are more capable of resource gain (Hobfoll et al., 2018) such as creative performance. On the other hand, resource gain spirals state that individuals with abundant resources are not only more able to acquire resources but also produce larger resource increments (Hobfoll et al., 2018). As a positive psychological resource, meaning of work can shape the intrinsic motivation and build enduring personal resources (e.g., energy, enthusiasm, and focus) (Allan et al., 2019). This helps to produce larger resource increments and in turn improve individual creative performance. Several empirical studies also demonstrated that meaning of work could increase individual creativity and creative behavior (Akgunduz et al., 2018; Liang et al., 2021). In summary, we believe that BSB indirectly affects creative performance through meaning of work. Based on these, we propose:

H2 Meaning of work mediates the relationship between BSB and employee creative performance.

The moderating role of team Chaxu climate

As shown in resource caravan passageways, individuals' creation and sustenance of the resources are influenced by ecological conditions, with organizations and the broader culture playing a major role in particular (Hobfoll et al., 2018). This means that the process of the boundary spanners acquiring resources through BSB is further influenced by organizational cultural factors. Compared to other factors, team Chaxu climate is a construct with Chinese indigenous features, summarizes the interpersonal relationship criteria of organizations in the context of Chinese culture, such as guanxi orientation, power hierarchy, differential treatment, and preferential resource allocation (Chen & Dian, 2018). It reflects the distribution characteristics of team resources and power (Liu et al., 2018). Therefore, we argue that team Chaxu climate may moderate the relationship between BSB and meaning of work.

Specifically, when team Chaxu climate is high, obvious circles and hierarchies will be formed around the center of power within the team, which will result in differences in team resource allocation (Liu et al., 2009). At this time, it is difficult for most outsiders to get care and support from their superiors or colleagues through BSB (Huang et al., 2018). They cannot fairly obtain rich resources from others in the process of resource investing (i.e., BSB), so they are not likely to perceive their value and significance (Wrzesniewski et al., 2003). At the same time, when individuals enhance their connection with leaders and colleagues through BSB, they can more obviously perceive the different treatment of leaders and the unfairness of interpersonal communication. They will produce negative emotions and negative self-perception, such as work pressure and self-doubt (Yang et al., 2021), thus consuming a lot of emotional and cognitive resources. This is not conducive to the formation of meaning of work. For a few insiders, while they can get more trust, care, support, and other resources from their leaders, they may be less willing to connect and share resources with their colleagues due to fear of losing their potential status and interests (Yang et al., 2021), and in turn reduce their motivation of boundary spanning, ultimately perceive less meaning of work. Furthermore, although members who perceive the high Chaxu climate can get support and relationships from external actors, which may make up for psychological imbalance caused by differential treatment of leaders to a certain extent. It may still reduce their commitment and obligation to the teams or leaders according to social exchange theory (Blau, 1964). Following this logic, we speculate that the positive perception of BSB may be weakened by the high team Chaxu climate, which reduces the meaning of work.

On the contrary, when team Chaxu climate is low, team leaders can treat their members fairly and the members are more likely to form workplace friendships. At this time, individuals can obtain rich resources from their superiors and colleagues through BSB to achieve team goals. This is conducive to the formation of meaning of work. Therefore, we argue that a low team Chaxu climate may strengthen the positive relationship between BSB and meaning of work. Therefore, we propose:

H3 Team Chaxu climate negatively moderates the relationship between BSB and meaning of work. That is, the team

Chaxu climate is lower, the positive relationship between BSB and meaning of work is stronger.

The moderated mediation effects

Combining hypothesis 2 and hypothesis 3, we argue that the team Chaxu climate moderates the indirect effect of meaning of work between BSB and employee creative performance. As mentioned above, it may limit the resource acquisition of boundary spanners and consume the cognitive, emotional, and other resources of individuals when the team Chaxu climate is higher. This reduces the meaning of work, ultimately leading to the decline of creative performance. In contrast, it may foster boundary spanners' resource acquisition when the Chaxu climate is lower. This enhances their meaning of work, ultimately leading to high creative performance. Accordingly, we propose:

H4 Team Chaxu climate negatively moderates the indirect effect of BSB on creative performance through meaning of work. That is, with the enhancement of team Chaxu climate, the indirect role of meaning of work between BSB and creative performance becomes lower.

Methodology

Procedure and samples

The research data of the field survey were collected from 6 companies located in Taiyuan, China. These companies involve the internet, landscape architecture, and consulting industries. In order to obtain the samples successfully, we first communicated with the company managers, explained the purpose of our research and asked for approval from them. Then, we identified a contact person at each company. They assisted the authors in identifying the team leaders and members (at least 4 people) who volunteered to participate in the survey and provided a list of participants. To ensure that the data can be effectively matched, the authors coded the team leaders and members on the list and asked them to fill in the code in the questionnaire. At the same time, to prevent the leakage of personal information, each member's code is known only to himself, the contact person and researchers. After fully communicating the requirements for filling in the questionnaire with the researchers, the contact person distributed questionnaires to 80 team leaders and 356 team members. All participants were informed that the data collection was only for academic purposes and the overall analysis was carried out without revealing

personal information. In the questionnaire, BSB, meaning of work, team Chaxu climate, and demographic variables were reported by team members, and team leaders evaluated the creative performance of team members. 211 valid samples from 46 teams were finally obtained after deleting invalid questionnaires, such as blank and regular answers, unmatched questionnaires, and samples with a team size of less than 3 people; the effective recovery rate is 59.27%. Each team leader rated 4.59 members on average. Among 211 employees, 40.3% were male, 57.3% had a bachelor's degree, 23.7% had a junior college degree, the average age was 29.62, and the average tenure was 4.37 years.

Measures

The scales were translated from English to Chinese through a standard back-translation method (Brislin, 1970). All measures used a five-point Likert scale (1 = strongly disagree to 5 = strongly agree).

BSB

Boundary spanning behavior was measured by a 6-item scale adopted by Marrone et al. (2007). A sample item was "persuade outsiders (e.g., faculty, clients) to support team decisions". Cronbach's alpha was 0.834 in this study.

Meaning of work

We used 6 items developed by May et al. (2004) to measure meaning of work. A sample item was "The work I do on this job is worthwhile". Cronbach's alpha was 0.859 in this study.

Employee creative performance

We measured employee creative performance with 3 items developed by Oldham and Cummings (1996). A sample item was "How original and practical is this person's work.? Original and practical work refers to developing ideas, methods, or products that are both totally unique and especially useful to the organization ". Cronbach's alpha was 0.873 in this study.

Team Chaxu climate

11 items developed by Liu (2003) were used to assess team Chaxu climate. A sample item was "my team leader has frequent contact with several subordinates". Cronbach's alpha was 0.937 in this study. This scale has good measurement validity for the Chinese (Liu et al., 2009). Because the variable was constructed on the team level, we used within-group

Table 1 CFA results of study variables

Model	χ^2	df	$\Delta\chi^2$	CFI	TLI	RMSEA
Four-factor model: BSB, MW, CX, CP	213.08	129	--	0.95	0.94	0.056
Three-factor model: BSB + MW, CX, CP	676.32	133	463.24**	0.70	0.61	0.139
Two-factor model: BSB + MW + CX, CP	1081.70	135	868.62**	0.47	0.33	0.183
One-factor model: BSB + MW + CX + CP	1146.81	136	933.73**	0.43	0.29	0.188

Note: ** $p < 0.01$; BSB = boundary spanning behavior, MW = meaning of work, CX = Chaxu climate at the individual level, CP = creative performance.

Table 2 Means, Standard deviations and Correlation among variables^a

Variables	M	SD	1	2	3	4	5	6	7
1. Gender	1.60	0.49							
2. Age	29.62	6.45	-0.10						
3. Education	2.57	0.79	-0.03	-0.49**					
4. Tenure	4.37	5.29	0.01	0.68**	-0.24**				
5. BSB	3.84	0.67	-0.06	0.11	-0.04	0.08			
6. MW	4.20	0.50	0.07	0.05	0.05	0.06	0.15*		
7. TCX ^b	3.41	0.58	0.05	-0.10	0.07	-0.17*	0.01	0.11	
8. CP	3.68	0.87	-0.09	0.07	-0.09	-0.01	0.62**	0.20**	0.17*

Note: * $p < 0.05$, ** $p < 0.01$, ^a $n = 221$ at the individual level, $N = 46$ at the team level, ^bThe team level variable is assigned to each individual in that team, BSB = boundary spanning behavior, MW = meaning of work, TCX = team Chaxu climate, CP = creative performance

reliability (ICC1), reliability of group mean (ICC2) and within-group agree indices (rwg) to judge whether team Chaxu climate of individuals reported is fit to be aggregated at the team level. Rwg, ICC1 and ICC2 of team Chaxu climate are 0.84, 0.35 and 0.71, respectively, which are above the threshold of 0.7, 0.12 and 0.47 in sequence.

Control variables

According to previous research (e.g., Liu et al., 2018; Marrone et al., 2007; Zhang & Li, 2020a), we controlled employee gender, age, education, and tenure at the individual level to eliminate the influence of these variables on the core variables in the present study. For gender, male = 1, female = 2; for age, measured in years old; for education, high school and below = 1, junior college degree = 2, bachelor degree = 3, master degree and above = 4; for tenure, measured in years.

Results

Preliminary analysis

We examined the common method variance (CMV) by Harman's single factor test (Podsakoff et al., 2003). By exploratory factor analysis (EFA), the first factor extracted without rotation does not exceed 40% of the total variation, indicating no serious common method bias. The results showed that the first factor accounted for 27.005% of the total variation in our study, which does not exceed

the above-recommended value. Due to the insensitivity of this method, this study further tested the CMV through the unmeasured latent method factor method (Podsakoff et al., 2003). We added one common factor to the four-factor model and contrasted the fit index. The results showed that χ^2 changed significantly ($\Delta\chi^2 = 53.26$, $\Delta df = 15$). Given that the sample size easily influenced it, we need to compare other fit indices. The five-factor model's other fit indices (CFI = 0.97, TLI = 0.96, RMSEA = 0.044) did not improve too much (not more than 0.02) (Williams et al., 1989). In summary, it can be concluded that the CMV is not serious in our study.

In addition, we conducted confirmatory factor analysis (CFA) by using Amos17.0 to test the discriminant vitality of individual-level variables, as shown in Table 1. The hypothesized four-factor model fit index ($\chi^2 = 213.08$, $df = 129$, CFI = 0.95, TLI = 0.94, RMSEA = 0.056) is better than other alternative models.

Descriptive statistics

The means, standard deviations, and correlation of variables in our study are shown in Table 2. BSB is both positively related to meaning of work ($\beta = 0.15$, $p < 0.05$) and creative performance ($\beta = 0.62$, $p < 0.01$). Meaning of work is positively related to creative performance ($\beta = 0.20$, $p < 0.01$).

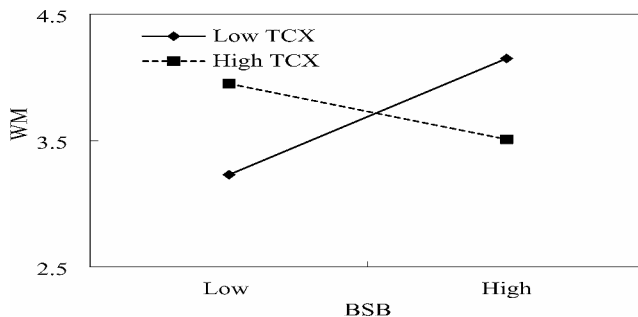
Testing of hypotheses

Our model is a two-level model; we used HLM6.08 to examine our hypotheses, as shown in Table 3. Before testing

Table 3 HLM results

Variables	WM				CP			
	Null model	M1	M2	M3	Null model	M4	M5	M6
Intercept	4.20***	3.00***	3.88***	3.71***	3.67***	3.83***	3.62***	3.76***
<i>Level 1</i>								
Gender		0.10	-0.08	0.09		-0.05	0.03	0
Age		0.01	0	0.01		0	0	0
Education		0.06	0.01	0.06		-0.04	-0.04	-0.04
Tenure		0	-0.01	0		-0.01	-0.01	-0.01
BSB			0.15*	0.16†			0.48***	0.46**
WM								0.22*
<i>Level 2</i>								
TCX				0.02				
<i>Level1×level2</i>								
BSB×TCX				-0.34*				
δ^2	0.22	0.22	0.20	0.25	0.45	0.46	0.36	0.35
τ_{00}	0.03	0.04	0.05	0	0.29	0.29	0.26	0.21

Note: † $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; $n = 221$ at individual level, $N = 46$ at team level

**Fig. 2** The moderating role of TCX between BSB and WM

the hypothesis, the data for level 1 was group-mean centered, and the data for level 2 was grand-mean centered. The results indicate that when we control gender, age, education, and tenure, BSB is positively related to creative performance (M5, $\gamma = 0.48$, $p < 0.001$). Therefore, Hypothesis 1 was supported.

We examined the mediating effect of meaning of work based on the three-step procedures proposed by Baron and Kenny (1986). First, the independent variable (BSB) is related to the dependent variable (CP), as supported by Hypothesis 1. Second, the independent variable (BSB) is related to the mediator (WM) ($\gamma = 0.15$, $p < 0.05$). Finally, when meaning of work was entered in the above model, the coefficient of BSB was reduced from 0.48 (M5, $p < 0.001$) to 0.46 (M6, $p < 0.01$) and the effect of meaning of work on employees' creative performance was significant ($\gamma = 0.22$, $p < 0.05$). The results suggest that meaning of work plays a partial mediating role between BSB and creative performance. Furthermore, we conducted the mediation test using the SPSS PROCESS. We found that the 95% confidence interval of the indirect effect was [0.03, 0.40], not including 0. These results show that the indirect effect of BSB

on employee creative performance via meaning of work is significant. Hence, Hypothesis 2 was supported.

The moderating role of team Chaxu climate was tested, as shown in Table 3. The interaction of BSB and team Chaxu climate (TCX) is significant (M3, $\gamma = -0.34$, $p < 0.05$). Therefore, Hypothesis 3 was initially supported. According to Aiken et al. (1991) suggestions, we further plotted the relationship between BSB and WM under the M+SD and M-SD of TCX, as shown in Fig. 2. The interaction plot shows that the positive relationship between BSB and WM is stronger when TCX is low (simple slope = 0.46, $p < 0.001$), while the relationship between BSB and WM is negative when TCX is high (simple slope = -0.22, $p < 0.05$). Combining both results, Hypothesis 3 was supported.

Finally, we tested hypothesis 4, that TCX moderated the mediating role of WM between BSB and CP by Mplus7.0. The results show that the confidence intervals of indirect effect contain 0 regardless of whether the TCX is high or low. That is, the moderated mediation effect does not exist. Hence, Hypothesis 4 was not supported.

Discussion

This study explored the influence mechanism of BSB on employee creative performance based on COR theory. The results find that BSB is positively related to creative performance, meaning of work mediates the relationship between BSB and creative performance, team Chaxu climate moderates the linkage between BSB and meaning of work. Still, the moderated mediation effect is not significant.

First, BSB is positively related to creative performance. This result indicates that BSB is an important independent variable of creative performance, which is consistent with

the findings of Zhang and Li (2020a). From the resource view, the main reason why BSB can improve creative performance is that BSB helps employees acquire various resources, such as relationships, influence, and heterogeneous knowledge, and in turn promotes individual creative performance. In addition, BSB may be considered as a resource-investing behavior. One exhibits BSB to get new resources (i.e., creative performance) and prevent them from falling into a resource loss spiral.

Second, meaning of work plays a partial mediating role between BSB and creative performance. It indicates that BSB affects creative performance directly and has an indirect influence on creative performance through meaning of work. Although there is no research exploring the mediating role of meaning of work between the above two variables, the empirical results support our speculation. According to COR theory, BSB is a resource-investing behavior that may help individuals acquire new resources, such as meaning of work. This resource is beneficial to increase employees' intrinsic motivation, engagement and positive emotions, which can improve their creative performance.

Third, team Chaxu climate plays a negative moderating role between BSB and meaning of work. The team Chaxu climate is stronger, the positive relationship between BSB and meaning of work is weaker. It is similar to the findings of many previous studies, such as Liu et al. (2009) and Yang et al. (2021). Specifically, employees who perceive a high team Chaxu climate are likely to feel marginalized by team leaders. They cannot get enough resources from leaders when they carry out boundary spanning activities. Meantime, antagonism, jealousy and distrust tend to be formed between team members, which is not conducive to gaining meaning of work.

Finally, the moderated mediation effects do not exist in our study. It demonstrates that team Chaxu climate does not moderate the indirect effect of meaning of work between BSB and creative performance. A possible explanation is that team Chaxu climate (moderates the mediating role of meaning of work) is constrained by other factors, such as team trust. Team trust refers to a shared perception that team members are willing to accept the possibility of being hurt based on positive expectations about the intentions or behaviors of other team members (Yang et al., 2021). It can reduce barriers between team members and form a more positive interpretation of the differential management behavior of leaders (Yang et al., 2021), which may statistically offset the negative influence of team Chaxu climate on meaning of work.

Theoretical implications

This study has some theoretical implications. First, this research is a response to the call to strengthen the research on the relationship between BSB and employee creative performance (Chen & Du, 2019). Unlike previous studies based on knowledge integration theory and status characteristics theory, this study answers the question of whether BSB affects employee creative performance based on COR theory, which provides a new insight to explain the relationship between them. It not only enriches the research on the relationship between BSB and employee creative performance but also expands the application of COR theory.

Second, we find that meaning of work mediates the linkage between BSB and creative performance, which answers how BSB affects creative performance. As mentioned above, the process of BSB affecting employee creative performance involves a dynamic change process of multiple resources. However, previous studies focusing on knowledge and informal status do not fully reveal the above processes. In contrast to previous studies, this study introduces the meaning of work as a mediating variable based on COR theory. It reveals the process that the resources obtained by individuals through BSB are transformed into creative performance within the individual. This enriches the research on the mediating mechanism of meaning of work.

Third, team Chaxu climate moderates the relationship between BSB and meaning of work, which reveals the boundary condition of the effectiveness of BSB in Chinese society. As pointed out by Liu et al. (2018), BSB originated in Western countries. We still know little about whether BSB plays an important role in non-western contexts, such as China. This research focuses on the local context in China and explores the moderating role of team Chaxu climate. The results show that the high team Chaxu climate weakened the positive influence of BSB on an individual's meaning of work. This finding provides a sample of empirical data regarding the effectiveness with which boundary spanning behavior can be implemented in the Chinese context, as well as expanding the research on team Chaxu climate, which contributes to a better understanding of China's complex modern society for scholars.

Practical implications

This research has some practical implications. First, team leaders should encourage members to exhibit BSB due to its positive influence on creative performance. For example, leaders can cultivate employees' awareness of crossing boundaries and encourage employees to actively collect information, seek feedback and obtain other resources from leaders, peers and external parties in the process of

completing tasks. At the same time, organizations should build bridges for employees to interact with internal and external parties. For example, organizations can increase the opportunities for employees to establish contact with external parties by holding expert lectures, cooperating with other enterprises.

Second, The mediating role of meaning of work indicates that employees' BSB improves creative performance by stimulating their meaning of work. Therefore, leaders should pay much attention to the meaning of work of members. For example, leaders should recognize and praise employees' efforts and achievements when employees carry out boundary spanning activities, thereby increasing employees' perception of the value and meaning of their work. Meanwhile, leaders can help employees learn more knowledge and develop new skills in the process of BSB by carrying out training activities, which can stimulate employees' interest and intrinsic motivation.

Finally, although Chaxu is a common phenomenon in China, team leaders should shape a climate of lower Chaxu to enhance the positive effects of BSB. For example, leaders should create a fair and equitable team atmosphere, such as reducing their behaviors of differential treatment and preferential resource allocation. At the same time, leaders should take measures to enhance communication and cooperation among team members, so as to form an equal and friendly interaction between the members. For example, enabling them to work together to complete a task.

Limitations and future research directions

There are some limitations to our study. First, although we constructed logical relationships between variables based on theory, we collected data through the paired surveys simultaneously. So this paper cannot draw a causal relationship between variables strictly. In particular, the relationship between meaning of work and BSB deserves further discussion. We acknowledge that if an employee has meaning of work, they may display more BSB. Therefore, in the future, experimental studies or longitudinal studies should be used. However, we still believe that BSB is more likely to affect meaning of work because BSB can enhance an individual's sense of value and importance, improve spanners' status roles, and develop high-quality relationships, which are the sources of meaning of work (Kahn, 1990). Meanwhile, we didn't control other key variables that may affect the relationship between BSB and employee creative performance. Future research can further refine this issue.

Second, we constructed the theoretical model based on COR theory and revealed the mechanism between BSB and creative performance. However, the partial mediating role of meaning of work reminds us that there may be other

mediators between BSB and creative performance. For instance, BSB may improve creative performance through team-member exchange (TMX) or informal status, according to the literature review of Marrone (2010). In addition, BSB may increase role overload. Thus, it is likely to have a "double-edged effect" on creative performance.

Finally, we only examined the moderating role of team Chaxu climate, other team contextual factors deserve to be investigated in the future. For example, as mentioned above, team trust may interact with the team Chaxu climate and further affect the linkage between BSB and meaning of work. Furthermore, team leadership and team task uncertainty (Liu et al., 2018) may be two possible moderating variables worthy of further testing in future studies.

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Data availability Some or all data, models, or codes generated or used during the study are available from the corresponding author by request.

Declarations

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. The study was approved by the Bioethics Committee of Shanxi University of Finance and Economics.

Consent to participate All participants gave informed consent.

Consent for publication We will grant the Publisher an exclusive license to publish the article.

Conflict of interest The author declares that there are no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

References

- Abukhait, R., Shamsudin, F. M., Bani-Melhem, S., & Al-Hawari, M. A. (2023). Obsessive-compulsive personality and creative performance: The moderating effect of manager coaching behavior. *Review of Managerial Science*, 17(1), 375–396. <https://doi.org/10.1007/s11846-022-00528-6>

- Aiken, L. S., West, S. G., & Reno, R. R. (1991). *Multiple regression: Testing and interpreting interactions*. Sage.
- Akgunduz, Y., Alkan, C., & Gök, Ö. A. (2018). Perceived organizational support, employee creativity and proactive personality: The mediating effect of meaning of work. *Journal of Hospitality and Tourism Management*, 34, 105–114. <https://doi.org/10.1016/j.jhtm.2018.01.004>
- Allan, B. A., Batz-Barbarich, C., Sterling, H. M., & Tay, L. (2019). Outcomes of meaningful work: A meta-analysis. *Journal of Management Studies*, 56(3), 500–528. <https://doi.org/10.1111/joms.12406>
- Bailey, C., Yeoman, R., Madden, A., Thompson, M., & Kerridge, G. (2019). A review of the empirical literature on meaningful work: Progress and research agenda. *Human Resource Development Review*, 18(1), 83–113. <https://doi.org/10.1177/1534484318804653>
- Baron, R., & Kenny, D. (1986). The moderator-mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations. *Journal of Personality and Social Psychology*, 51(6), 1173–1182. <https://doi.org/10.1037/0022-3514.51.6.1173>
- Blau, P. M. (1964). *Exchange and Power in Social Life*. Wiley.
- Brislin, R. W. (1970). Back-translation for cross-cultural research. *Journal of Cross-Cultural Psychology*, 1(3), 185–216. <https://doi.org/10.1177/135910457000100301>
- Chen, Z. X., & Dian, Y. J. (2018). Organizational Chaxu climate: Concept, measurement and action mechanism. *Foreign Economy and Management*, 40(6), 86–98. <https://doi.org/10.16538/j.cnki.fem.2018.06.007>
- Chen, Y., & Du, P. C. (2019). Research focuses and frontiers of boundary-spanning behavior in the organization field—Based on the analysis of mapping knowledge domains. *Science & Technology Progress and Policy*, 36(9), 151–159. <https://doi.org/10.6049/kjbydc.L201808538>
- Clausen, T., & Borg, V. (2011). Job demands, job resources and meaning at work. *Journal of Managerial Psychology*, 26(8), 665–681. <https://doi.org/10.1108/02683941111181761>
- Cohen-Meitar, R., Carmeli, A., & Waldman, D. A. (2009). Linking meaningfulness in the workplace to employee creativity: The intervening role of organizational identification and positive psychological experiences. *Creativity Research Journal*, 21(4), 361–375. <https://doi.org/10.1080/10400410902969910>
- Duan, J., Wang, X., Liu, Y., & Han, L. (2023). Linking paternalistic leadership to employee creativity: A meaning-based perspective. *Leadership & Organization Development Journal*, ahead-of-print. <https://doi.org/10.1108/LODJ-11-2021-0497>
- Hargadon, A. B. (1998). Firms as knowledge brokers: Lessons in pursuing continuous innovation. *California Management Review*, 40(3), 209–227. <https://doi.org/10.2307/41165951>
- Hobfoll, S. E. (1989). Conservation of resources: A new attempt at conceptualizing stress. *American Psychologist*, 44(3), 513–524. <https://doi.org/10.1037/0003-066x.44.3.513>
- Hobfoll, S. E. (2011). Conservation of resource caravans and engaged settings. *Journal of Occupational and Organizational Psychology*, 84(1), 116–122. <https://doi.org/10.1111/j.2044-8325.2010.02016.x>
- Hobfoll, S. E., Halbesleben, J., Neveu, J. P., & Westman, M. (2018). Conservation of resources in the organizational context: The reality of resources and their consequences. *Annual Review of Organizational Psychology and Organizational Behavior*, 5(1), 103–128. <https://doi.org/10.1146/annurev-orgpsych-032117-104640>
- Huang, Y. L., Wang, Y., Wei, Z. B., & Liu, Z. Y. (2018). The impact of climate of Chaxu on career calling: The role of self-esteem and psychological empowerment. *Human Resources Development of China*, 35(9), 16–28. <https://doi.org/10.16471/j.cnki.11-2822/c.2018.09.002>
- Kahn, W. A. (1990). Psychological conditions of personal engagement and disengagement at work. *Academy of Management Journal*, 33(4), 692–724. <https://doi.org/10.2307/256287>
- Liang, B. Q., Van Knippenberg, D., & Gu, Q. X. (2021). A cross-level model of shared leadership, meaning, and individual creativity. *Journal of Organizational Behavior*, 42(1), 68–83. <https://doi.org/10.1002/job.2494>
- Liao, H., Liu, D., & Loi, R. (2010). Looking at both sides of the social exchange coin: A social cognitive perspective on the joint effects of relationship quality and differentiation on creativity. *Academy of Management Journal*, 53(5), 1090–1109. <https://doi.org/10.5465/amj.2010.54533207>
- Liao, H. H., Huang, L., & Hu, B. (2022). Conservation of resources theory in the organizational behavior context: Theoretical evolution and challenges. *Advances in Psychological Science*, 30(2), 449–463.
- Liu, Z. S. (2003). *The influence of Chaxu climate on the subordinates' working attitudes and their behaviors* Unpublished master's thesis, Tung Wah University, Taiwan.
- Liu, J., Zhang, K., & Zhong, L. F. (2009). The formation and impact of the atmosphere of the error routine of the work team: A case study based on successive data. *Management World*, 8(188), 92–101. <https://doi.org/10.19744/j.cnki.11-1235/f.2009.08.011>
- Liu, S., Schuler, R., & Zhang, P. (2013). External learning activities and employee creativity in Chinese R&D teams. *Cross Cultural Management: An International Journal*, 20, 429–448. <https://doi.org/10.1108/CCM-10-2012-0088>
- Liu, S. B., Jiang, K. F., Chen, J. X., Pan, J. Z., & Lin, X. S. (2018). Linking employee boundary spanning behavior to task performance: The influence of informal leader emergence and group power distance. *International Journal of Human Resource Management*, 29(12), 1879–1899. <https://doi.org/10.1080/09585192.2016.1216872>
- Lysova, E. I., Allan, B. A., Dik, B. J., Duffy, R. D., & Steger, M. F. (2019). Fostering meaningful work in organizations: A multi-level review and integration. *Journal of Vocational Behavior*, 110, 374–389. <https://doi.org/10.1016/j.jvb.2018.07.004>
- Marrone, J. A. (2010). Team boundary spanning: A multilevel review of past research and proposals for the future. *Journal of Management*, 36(4), 911–940. <https://doi.org/10.1177/0149206309353945>
- Marrone, J. A., Tesluk, P. E., & Carson, J. B. (2007). A multilevel investigation of antecedents and consequences of team member boundary-spanning behavior. *Academy of Management Journal*, 50(6), 1423–1439. <https://doi.org/10.5465/amj.2007.28225967>
- May, D. R., Gilson, R. L., & Harter, L. M. (2004). The psychological conditions of meaningfulness, safety and availability and the engagement of the human spirit at work. *Journal of Occupational and Organizational Psychology*, 77(1), 11–37. <https://doi.org/10.1348/096317904322915892>
- Oldham, G. R., & Cummings, A. (1996). Employee creativity: Personal and contextual factors at work. *Academy of Management Journal*, 39(3), 607–634. <https://doi.org/10.2307/256657>
- Podsakoff, P. M., MacKenzie, S. B., Lee, J. Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology*, 88(5), 879–903. <https://doi.org/10.1037/0021-9010.88.5.879>
- Rosso, B. D., Dekas, K. H., & Wrzesniewski, A. (2010). On the meaning of work: A theoretical integration and review. *Research in Organizational Behavior*, 30, 91–127. <https://doi.org/10.1016/j.riob.2010.09.001>
- Shalley, C. E., Zhou, J., & Oldham, G. R. (2004). The effects of personal and contextual characteristics on creativity: Where should we go from here. *Journal of Management*, 30(6), 933–958. <https://doi.org/10.1016/j.jm.2004.06.007>

- Song, M., Huang, Z. J., Hu, H. Y., & Qi, M. (2018). Literature review and future prospect on work meaningfulness. *Human Resources Development of China*, 35(9), 85–96. <https://doi.org/10.16471/j.cnki.11-2822/c.2018.09.008>
- Song, M., Hu, H. Y., & Wang, Z. (2021). Benefits or costs? The positive and negative effects of leader boundary spanning behavior on leader performance. *Management Review*, 33(4), 236–247. <https://doi.org/10.14120/j.cnki.cn11-5057/f.2021.04.020>
- Sun, J., Lee, J. W., & Sohn, Y. W. (2019). Work context and turnover intention in social enterprises: The mediating role of meaning of work. *Journal of Managerial Psychology*, 34(1), 46–60. <https://doi.org/10.1108/jmp-11-2017-0412>
- Wadei, K. A., Asaah, J. A., Amoah-Ashyiah, A., & Wadei, B. (2023). Do the right thing the right way! How ethical leaders increase employees creative performance. *Journal of Leadership & Organizational Studies*, 30(4), 428–441. <https://doi.org/10.1177/15480518231195597>
- Wang, H. Y., & Kou, X. L. (2023). The impact of error management climate to bootlegging: A moderated mediation model. *Journal of Technology Economics*, 42(1), 53–63.
- Williams, L. J., Cote, J. A., & Buckley, M. R. (1989). Lack of method variance in self-reported affect and perceptions at work: Reality or artifact. *Journal of Applied Psychology*, 74(3), 462–468. <https://doi.org/10.1037/0021-9010.74.3.462>
- Wrzesniewski, A. C., Dutton, J. E., & Debebe, G. (2003). Interpersonal sensemaking and the meaning of work. *Research in Organizational Behavior*, 25(2), 93–135. [https://doi.org/10.1016/s0191-3085\(03\)25003-6](https://doi.org/10.1016/s0191-3085(03)25003-6)
- Yang, H., Peng, C., Du, G., Xu, C., & Lv, C. L. (2021). Ambidextrous leadership, team differential atmosphere and the creativity of R&D team. *Scientific Research*, 39(7), 1248–1256. <https://doi.org/10.16192/j.cnki.1003-2053.20210129.007>
- Yuan, F. R., & Woodman, R. W. (2010). Innovative behavior in the workplace: The role of performance and image outcome expectations. *Academy of Management Journal*, 53(2), 323–342. <https://doi.org/10.5465/amj.2010.49388995>
- Zhang, Q., & Li, J. (2020a). Can employee's boundary-spanning behavior exactly promote innovation performance? The roles of creative ideas generation and team task interdependence. *International Journal of Manpower*, 42(6), 1047–1063. <https://doi.org/10.1108/IJM-06-2019-0302>
- Zhang, Z., & Li, R. (2023b). How and when employees' boundary-spanning behavior improves their creative performance: A moderated mediation model. *Journal of Management & Organization*, 1–17. <https://doi.org/10.1017/jmo.2023.51>

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